

Introduction to Version Control: The Digital Time Machine

Grade 6 | Session 1

Welcome, Code Explorers! Have you ever worked on a school project for hours, only to accidentally delete it? Or maybe you made a change you hated and couldn't go back? Version Control is here to save the day. Think of it as a professional 'Undo' button for everything you create.

Explorer Tip: Always remember that in Git, nothing is ever truly lost. It's your safety net for coding!

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The Problem: File Name Chaos

Why standard saving isn't enough

We've all seen files named 'Project_Final.doc', 'Project_Final_v2.doc', and 'Project_REAL_FINAL_I_PROMISE.doc'. This is messy! It makes it impossible to know which version is actually the latest. Version control replaces this chaos with a clean, organized history of your work.

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What is Git?

Your Personal Digital Notebook

Git is a tool you install on your computer. It acts like a personal diary for your code. Every time you finish a small task, you tell Git to take a 'snapshot' of your work. This way, if you make a mistake tomorrow, you can simply jump back to the snapshot you took today.

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What is GitHub?

The National Library of India for Code

If Git is your private notebook, GitHub is like a massive public library. It's an online platform where you can 'push' your work so others can see it, help you, or work on it with you. Millions of developers in India and across the world use it to build the apps we use every day.

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Key Term: The Repository

Your Digital Almirah

A Repository (or 'Repo') is like a special folder where Git tracks everything. Think of it as your digital Almirah or school locker. Everything inside this locker is safely backed up and organized by time and date.

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Key Term: The Commit

Saving with a Purpose

A 'Commit' is a snapshot of your project at a specific point in time. Unlike a regular 'Save', a commit always comes with a note called a 'Commit Message'. This message explains **what** you changed so your future self knows exactly what happened.

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Activity: Human Git

Classroom Exercise

Let's practice! Pass a single sheet of paper around the room. Each student adds one sentence to a story. Before passing it, you must write a 'Commit Message' on a sticky note (e.g., 'Added a dragon to the story'). This is exactly how developers track changes!

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Writing Good Commit Messages

The Art of Clear Notes

Bad messages like 'abc' or 'fixed' are unhelpful. Good messages tell a story. Instead of 'stuff', write 'Changed background to blue'. Clear messages help your teammates (and your future self!) understand your logic.

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Indian Context: ISRO

Version Control in High Stakes

Scientists at ISRO use version control for every rocket launch. Imagine if they lost the code for a satellite mid-flight! They track every single version of their software to ensure that missions like Chandrayaan are successful and safe.

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Branching: Parallel Worlds

Exploring Different Paths

Sometimes you want to try a new idea without breaking your main project. 'Branching' lets you create a separate path. It's like a 'choose your own adventure' book where you can explore one ending, then come back and try a different one.

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Recap & Success Criteria

Are you a Version Control Hero?

Today we learned that: 1. Version Control is a Time Machine. 2. Git is for you; GitHub is for the world. 3. Commits are milestones with notes. 4. Repositories are your digital lockers. You're now ready to start your journey as a Code Explorer!

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